

References

- Abdel-Hamid, Ossama, Abdel-rahman Mohamed, Hui Jiang, Li Deng, Gerald Penn, and Dong Yu. 2014. "Convolutional Neural Networks for Speech Recognition." *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 22 (10): 1533–45. <https://doi.org/10.1109/TASLP.2014.2339736>.
- Alqahtani, Hamed, Manolya Kavakli-Thorne, and Gulshan Kumar. 2019. "Applications of Generative Adversarial Networks (GANs): An Updated Review." *Archives of Computational Methods in Engineering*, December. <https://doi.org/10.1007/s11831-019-09388-y>.
- Ang, Andrew, Robert J Hodrick, Yuhang Xing, and Xiaoyan Zhang. 2006. "The Cross-Section of Volatility and Expected Returns." *The Journal of Finance* 61 (1): 259–99.
- Ang, Andrew. 2014. *Asset Management: A Systematic Approach to Factor Investing*. 1 edition. Oxford: Oxford University Press.
- Araci, Dogu. 2019. "FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models." *ArXiv:1908.10063 [Cs]*, August. <http://arxiv.org/abs/1908.10063>.
- Arora, Saurabh, and Prashant Doshi. 2019. "A Survey of Inverse Reinforcement Learning: Challenges, Methods and Progress." *ArXiv:1806.06877 [Cs, Stat]*, August. <http://arxiv.org/abs/1806.06877>.
- Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen. 2013. "Value and Momentum Everywhere." *The Journal of Finance* 68 (3): 929–85. <https://www.jstor.org/stable/42002613>.
- Bahdanau, Dzmitry, Kyunghyun Cho, and Yoshua Bengio. 2016. "Neural Machine Translation by Jointly Learning to Align and Translate." *ArXiv:1409.0473 [Cs, Stat]*, May. <http://arxiv.org/abs/1409.0473>.
- Bailey, David H., Jonathan M. Borwein, and Marcos Lopez de Prado. 2016. *The Probability of Backtest Overfitting*. *Journal of Computational Finance*, September. <https://www.risk.net/node/2471206>.
- Banz, Rolf W. 1981. "The Relationship between Return and Market Value of Common Stocks." *Journal of Financial Economics* 9 (1): 3–18.

- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny. 1998. "A Model of Investor Sentiment." *Journal of Financial Economics* 49 (3): 307–43.
- Basu, Sanjoy, and others. 1981. "The Relationship between Earnings' Yield, Market Value and Return for NYSE Common Stocks: Further Evidence."
- Bengio, Y., A. Courville, and P. Vincent. 2013. "Representation Learning: A Review and New Perspectives." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35 (8): 1798–1828. <https://doi.org/10.1109/TPAMI.2013.50>.
- Betancourt, Michael. 2018. "A Conceptual Introduction to Hamiltonian Monte Carlo." *ArXiv:1701.02434 [Stat]*, July. <http://arxiv.org/abs/1701.02434>.
- Bishop, Christopher. 2006. *Pattern Recognition and Machine Learning*. Information Science and Statistics. New York: Springer-Verlag. <https://www.springer.com/gp/book/9780387310732>.
- Blei, David M., Andrew Y. Ng, and Michael I. Jordan. 2003. "Latent Dirichlet Allocation." *Journal of Machine Learning Research* 3 (Jan): 993–1022. <http://jmlr.csail.mit.edu/papers/v3/blei03a.html>.
- Burges, Chris J. C. 2010. "Dimension Reduction: A Guided Tour." *Foundations and Trends in Machine Learning*, January. <https://www.microsoft.com/en-us/research/publication/dimension-reduction-a-guided-tour-2/>.
- Byrd, David, Maria Hybinette, and Tucker Hybinette Balch. 2019. "ABIDES: Towards High-Fidelity Market Simulation for AI Research." *ArXiv:1904.12066 [Cs]*, April. <http://arxiv.org/abs/1904.12066>.
- Casella, George, and Edward I. George. 1992. "Explaining the Gibbs Sampler." *The American Statistician* 46 (3): 167–74. <https://doi.org/10.2307/2685208>.
- Chan, Ernie., 2008. *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*, 1 edition. ed. Wiley, Hoboken, N.J.
- Chan, Ernie. 2013. *Algorithmic Trading: Winning Strategies and Their Rationale*. 1st ed. Wiley Publishing.
- Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16*, 785–94. <https://doi.org/10.1145/2939672.2939785>.
- Chen, Wei, Tie-yan Liu, Yanyan Lan, Zhi-ming Ma, and Hang Li. 2009. "Ranking Measures and Loss Functions in Learning to Rank." In *Advances in Neural Information Processing Systems 22*, edited by Y. Bengio, D. Schuurmans, J. D. Lafferty, C. K. I. Williams, and A. Culotta, 315–323. Curran Associates, Inc. <http://papers.nips.cc/paper/3708-ranking-measures-and-loss-functions-in-learning-to-rank.pdf>.
- Cheung, W., 2010. *The Black-Litterman model explained*. *J Asset Manag* 11, 229–243. <https://doi.org/10.1057/jam.2009.28>.
- Chib, Siddhartha, and Edward Greenberg. 1995. "Understanding the Metropolis-Hastings Algorithm." *The American Statistician* 49 (4): 327–35. <https://doi.org/10.1080/00031305.1995.10476177>.

- Cho, Kyunghyun, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. 2014. "Learning Phrase Representations Using RNN Encoder-Decoder for Statistical Machine Translation." *In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1724–1734. Doha, Qatar: Association for Computational Linguistics. <https://doi.org/10.3115/v1/D14-1179>.
- Chung, Junyoung, Caglar Gulcehre, Kyunghyun Cho, and Yoshua Bengio. 2014. "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling." *NIPS 2014 Workshop on Deep Learning*, December 2014. <https://nyuscholars.nyu.edu/en/publications/empirical-evaluation-of-gated-recurrent-neural-networks-on-sequen>.
- Clarke, R., Silva, H. de, Thorley, S., 2002. *Portfolio Constraints and the Fundamental Law of Active Management*. Financial Analysts Journal 58, 48–66.
- Creswell, Antonia, Tom White, Vincent Dumoulin, Kai Arulkumaran, Biswa Sengupta, and Anil A. Bharath. 2018. "Generative Adversarial Networks: An Overview." *IEEE Signal Processing Magazine* 35 (1): 53–65. <https://doi.org/10.1109/MSP.2017.2765202>.
- Cubuk, Ekin D. 2019. "AutoAugment: Learning Augmentation Strategies From Data." *CVFPR*, 11.
- Cummins, Mark, and Andrea Bucca. 2012. "Quantitative Spread Trading on Crude Oil and Refined Products Markets." *Quantitative Finance* 12 (12): 1857–75. <https://doi.org/10.1080/14697688.2012.715749>.
- Cybenko, G. 1989. "Approximation by Superpositions of a Sigmoidal Function." *Mathematics of Control, Signals and Systems* 2 (4): 303–14. <https://doi.org/10.1007/BF02551274>.
- David H. Bailey et al. (2015), *Backtest Overfitting: An Interactive Example*. <http://datagrid.lbl.gov/backtest/>.
- De Prado, Marcos Lopez. 2018. *Advances in Financial Machine Learning*. John Wiley & Sons.
- DeMiguel, V., Garlappi, L., Uppal, R., 2009. *Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?* Rev Financ Stud 22, 1915–1953. <https://doi.org/10.1093/rfs/hhm075>.
- Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding." *In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 4171–4186. Minneapolis, Minnesota: Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>.
- Dumais, S. T., G. W. Furnas, T. K. Landauer, S. Deerwester, and R. Harshman. 1988. "Using Latent Semantic Analysis to Improve Access to Textual Information." *In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 281–285. CHI '88. Washington, D.C., USA: Association for Computing Machinery. <https://doi.org/10.1145/57167.57214>.

- Elliott, Robert J., John Van Der Hoek, and William P. Malcolm. 2005. "Pairs Trading." *Quantitative Finance* 5 (3): 271–76. <https://doi.org/10.1080/14697680500149370>.
- Esposito, F., D. Malerba, G. Semeraro, and J. Kay. 1997. "A Comparative Analysis of Methods for Pruning Decision Trees." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 19 (5): 476–91. <https://doi.org/10.1109/34.589207>.
- Esteban, Cristóbal, Stephanie L. Hyland, and Gunnar Rätsch. 2017. "Real-Valued (Medical) Time Series Generation with Recurrent Conditional GANs." *ArXiv:1706.02633 [Cs, Stat]*, December. <http://arxiv.org/abs/1706.02633>.
- Fabozzi, Frank J, Sergio M Focardi, and Petter N Kolm. 2010. *Quantitative Equity Investing: Techniques and Strategies*. John Wiley & Sons.
- Fama, E.F., French, K.R., 2004. *The Capital Asset Pricing Model: Theory and Evidence*. *Journal of Economic Perspectives* 18, 25–46. <https://doi.org/10.1257/0895330042162430>.
- Fama, Eugene F, and James D MacBeth. 1973. "Risk, Return, and Equilibrium: Empirical Tests." *Journal of Political Economy* 81 (3): 607–36.
- Fama, Eugene F, and Kenneth R French. 1993. "Common Risk Factors in the Returns on Stocks and Bonds." *Journal of Financial Economics* 33: 3–56.
- Fama, Eugene F, and Kenneth R French. 1998. "Value versus Growth: The *International Evidence*." *The Journal of Finance* 53 (6): 1975–99.
- Fama, Eugene F., and Kenneth R. French. 2015. "A Five-Factor Asset Pricing Model." *Journal of Financial Economics* 116 (1): 1–22. <https://doi.org/10.1016/j.jfineco.2014.10.010>.
- Fawcett, Tom. 2006. "An Introduction to ROC Analysis." *Pattern Recognition Letters, ROC Analysis in Pattern Recognition*, 27 (8): 861–74. <https://doi.org/10.1016/j.patrec.2005.10.010>.
- Fefferman, Charles, Sanjoy Mitter, and Hariharan Narayanan. 2016. "Testing the Manifold Hypothesis." *Journal of the American Mathematical Society* 29 (4): 983–1049. <https://doi.org/10.1090/jams/852>.
- Fei-Fei, Li. 2015. "ImageNet Large Scale Visual Recognition Challenge." *International Journal of Computer Vision* 115 (3): 211–52. <https://doi.org/10.1007/s11263-015-0816-y>.
- Fisher, Walter D. 1958. "On Grouping for Maximum Homogeneity." *Journal of the American Statistical Association* 53 (284): 789–98. <https://doi.org/10.2307/2281952>.
- Freund, Yoav, and Robert E Schapire. 1997. "A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting." *Journal of Computer and System Sciences* 55 (1): 119–39. <https://doi.org/10.1006/jcss.1997.1504>.
- Friedman, Jerome H. 2001. "Greedy Function Approximation: A Gradient Boosting Machine." *The Annals of Statistics* 29 (5): 1189–1232. <https://www.jstor.org/stable/2699986>.

- Fu, Rao, Jie Chen, Shutian Zeng, Yiping Zhuang, and Agus Sudjianto. 2019. "Time Series Simulation by Conditional Generative Adversarial Net." *ArXiv:1904.11419 [Cs, Eess, Stat]*, April. <http://arxiv.org/abs/1904.11419>.
- Gatev, Evan, William N. Goetzmann, and K. Geert Rouwenhorst. 2006. "Pairs Trading: Performance of a Relative-Value Arbitrage Rule." *The Review of Financial Studies* 19 (3): 797–827. <https://doi.org/10.1093/rfs/hhj020>.
- Gelman, Andrew, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. 2013. *Bayesian Data Analysis, Third Edition*. CRC Press.
- Gómez, David, and Alfonso Rojas. 2015. "An Empirical Overview of the No Free Lunch Theorem and Its Effect on Real-World Machine Learning Classification." *Neural Computation* 28 (1): 216–28. https://doi.org/10.1162/NECO_a_00793.
- Gonzalo, Jesús, and Tae-Hwy Lee. 1998. "Pitfalls in Testing for Long-Run Relationships." *Journal of Econometrics* 86 (1): 129–54. [https://doi.org/10.1016/S0304-4076\(97\)00111-5](https://doi.org/10.1016/S0304-4076(97)00111-5).
- Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. "Generative Adversarial Nets." In *Advances in Neural Information Processing Systems 27*, edited by Z. Ghahramani, M. Welling, C. Cortes, N. D. Lawrence, and K. Q. Weinberger, 2672–2680. Curran Associates, Inc. <http://papers.nips.cc/paper/5423-generative-adversarial-nets.pdf>.
- Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. 2016. *Deep Learning*. MIT press.
- Goodfellow, Ian. 2014. "Multi-Digit Number Recognition from Street View Imagery Using Deep Convolutional Neural Networks." In *ICLR2014*.
- Goyal, Amit. 2012. "Empirical Cross-Sectional Asset Pricing: A Survey." *Financial Markets and Portfolio Management* 26 (1): 3–38. <https://doi.org/10.1007/s11408-011-0177-7>.
- Graham, Benjamin, David Dodd, and David Le Fevre Dodd. 1934. *Security Analysis: The Classic 1934 Edition*. McGraw Hill Professional.
- Green, Jeremiah, John R. M. Hand, and X. Frank Zhang. 2017. "The Characteristics That Provide Independent Information about Average U.S. Monthly Stock Returns." *The Review of Financial Studies* 30 (12): 4389–4436. <https://doi.org/10.1093/rfs/hhx019>.
- Grinold, R.C., 1989. *The fundamental law of active management*. The Journal of Portfolio Management 15, 30–37. <https://doi.org/10.3905/jpm.1989.409211>.
- Gu, S., Kelly, B., and Xu, D. 2020 "Autoencoder Asset Pricing Models." *Journal of Econometrics* (forthcoming).
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu. 2020. "Empirical Asset Pricing via Machine Learning." *The Review of Financial Studies*, no. hhaa009 (February). <https://doi.org/10.1093/rfs/hhaa009>.
- Harris, Larry. 2003. *Trading and Exchanges: Market Microstructure for Practitioners*. Oxford University Press.

- Hasselt, Hado van, Arthur Guez, and David Silver. 2015. "Deep Reinforcement Learning with Double Q-Learning." *ArXiv:1509.06461 [Cs]*, September. <http://arxiv.org/abs/1509.06461>.
- Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition*. 2nd ed. Springer Series in Statistics. New York: Springer-Verlag. <https://doi.org/10.1007/978-0-387-84858-7>.
- Hastie, Trevor, Robert Tibshirani, and Martin Wainwright. 2015. *Statistical Learning with Sparsity: The Lasso and Generalizations*. CRC press.
- He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2015. "Deep Residual Learning for Image Recognition." *ArXiv:1512.03385 [Cs]*, December. <http://arxiv.org/abs/1512.03385>.
- Hendricks, Dieter, and Diane Wilcox. 2014. "A Reinforcement Learning Extension to the Almgren-Chriss Framework for Optimal Trade Execution." In *2014 IEEE Conference on Computational Intelligence for Financial Engineering Economics (CIFER)*, 457–64. <https://doi.org/10.1109/CIFER.2014.6924109>.
- Hessel, Matteo, Joseph Modayil, Hado van Hasselt, Tom Schaul, Georg Ostrovski, Will Dabney, Dan Horgan, Bilal Piot, Mohammad Azar, and David Silver. 2017. "Rainbow: Combining Improvements in Deep Reinforcement Learning." *ArXiv:1710.02298 [Cs]*, October. <http://arxiv.org/abs/1710.02298>.
- Hihi, Salah El, and Yoshua Bengio. 1996. "Hierarchical Recurrent Neural Networks for Long-Term Dependencies." In *Advances in Neural Information Processing Systems 8*, edited by D. S. Touretzky, M. C. Mozer, and M. E. Hasselmo, 493–499. MIT Press. <http://papers.nips.cc/paper/1102-hierarchical-recurrent-neural-networks-for-long-term-dependencies.pdf>.
- Hochreiter, Sepp, and Jürgen Schmidhuber. 1996. "LSTM Can Solve Hard Long Time Lag Problems." In *Proceedings of the 9th International Conference on Neural Information Processing Systems*, 473–479. NIPS'96. Denver, Colorado: MIT Press.
- Hochreiter, Sepp, Yoshua Bengio, Paolo Frasconi, Jürgen Schmidhuber, and others. 2001. "Gradient Flow in Recurrent Nets: The Difficulty of Learning Long-Term Dependencies."
- Hoerl, Arthur E, and Robert W Kennard. 1970. "Ridge Regression: Biased Estimation for Nonorthogonal Problems." *Technometrics* 12 (1): 55–67.
- Hoffman, Matthew D., and Andrew Gelman. 2011. "The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo." *ArXiv:1111.4246 [Cs, Stat]*, November. <http://arxiv.org/abs/1111.4246>.
- Hofmann, Thomas. 2001. "Unsupervised Learning by Probabilistic Latent Semantic Analysis." *Machine Learning* 42 (1): 177–96. <https://doi.org/10.1023/A:1007617005950>.
- Hong, Harrison, Terence Lim, and Jeremy C. Stein. 2000. "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies." *The Journal of Finance* 55 (1): 265–95. <https://doi.org/10.1111/0022-1082.00206>.

- Hornik, Kurt. 1991. "Approximation Capabilities of Multilayer Feedforward Networks." *Neural Networks* 4 (2): 251–57. [https://doi.org/10.1016/0893-6080\(91\)90009-T](https://doi.org/10.1016/0893-6080(91)90009-T).
- Hou, Kewei, Chen Xue, and Lu Zhang. 2015. "Digesting Anomalies: An Investment Approach." *The Review of Financial Studies* 28 (3): 650–705. <https://doi.org/10.1093/rfs/hhu068>.
- Hou, K., Xue, C., Zhang, L., 2017. *Replicating Anomalies* (SSRN Scholarly Paper No. ID 2961979). Social Science Research Network, Rochester, NY.
- Huang, Gao, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. 2018. "Densely Connected Convolutional Networks." *ArXiv:1608.06993 [Cs]*, January. <http://arxiv.org/abs/1608.06993>.
- Ismail Fawaz, Hassan, Germain Forestier, Jonathan Weber, Lhassane Idoumghar, and Pierre-Alain Muller. 2019. "Deep Learning for Time Series Classification: A Review." *Data Mining and Knowledge Discovery* 33 (4): 917–63. <https://doi.org/10.1007/s10618-019-00619-1>.
- Jaeger, Herbert. 2001. "The 'Echo State' Approach to Analysing and Training Recurrent Neural Networks with an Erratum Note." *Bonn, Germany: German National Research Center for Information Technology GMD Technical Report* 148 (34): 13.
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2013. *An Introduction to Statistical Learning*. Vol. 112. Springer.
- Jegadeesh, Narasimhan, and Sheridan Titman. 1993. "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." *The Journal of Finance* 48 (1): 65–91.
- Jones, Charles. 2018. "Understanding the Market for Us Equity Market Data." NYSE. <https://www0.gsb.columbia.edu/faculty/cjones/papers/2018.08.31%20US%20Equity%20Market%20Data%20Paper.pdf>.
- JP Morgan, 2012. *Improving on risk parity – Hedging Forecast Uncertainty*. https://am.jpmorgan.com/blobcontent/1378404528937/83456/11_77%20Risk%20Parity.pdf.
- Ke, Guolin, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." In *Advances in Neural Information Processing Systems* 30, edited by I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, 3146–3154. Curran Associates, Inc. <http://papers.nips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree.pdf>.
- Kearns, Michael, and Yuriy Nevmyvaka. 2013. "Machine Learning for Market Microstructure and High Frequency Trading." *High Frequency Trading: New Realities for Traders, Markets, and Regulators*.
- Kelley, David. 2019. "Which Leading Indicators Have Done Better at Signaling Past Recessions?" *Chicago Fed Letter*, no. 425: 1.
- Kelly, Bryan T., Seth Pruitt, and Yinan Su. 2019. "Characteristics Are Covariances: A Unified Model of Risk and Return." *Journal of Financial Economics* 134 (3): 501–24. <https://doi.org/10.1016/j.jfineco.2019.05.001>.

- Kelly, J.L., 2011. *A New Interpretation of Information Rate*, in: The Kelly Capital Growth Investment Criterion, World Scientific Handbook in Financial Economics Series. WORLD SCIENTIFIC, pp. 25–34. https://doi.org/10.1142/9789814293501_0003.
- Kingma, Diederik P., and Max Welling. 2014. "Auto-Encoding Variational Bayes." *ArXiv:1312.6114 [Cs, Stat]*, May. <http://arxiv.org/abs/1312.6114>.
- Kingma, Diederik, and Jimmy Ba. 2014. "Adam: A Method for Stochastic Optimization," December. <https://arxiv.org/abs/1412.6980v8>.
- Kingma, Diederik P., and Max Welling. 2019. "An Introduction to Variational Autoencoders." *Foundations and Trends® in Machine Learning* 12 (4): 307–92. <https://doi.org/10.1561/22000000056>.
- Kolanovic, Marko, and Rajesh Krishnamachari. 2017. "Big Data and AI Strategies - Machine Learning and Alternative Data Approach to Investing." White Paper. JP Morgan. <http://www.fullertreacymoney.com/system/data/files/PDFs/2017/October/18th/Big%20Data%20and%20AI%20Strategies%20-%20Machine%20Learning%20and%20Alternative%20Data%20Approach%20to%20Investing.pdf>.
- Koshiyama, Adriano, Nick Firoozye, and Philip Treleaven. 2019. "Generative Adversarial Networks for Financial Trading Strategies Fine-Tuning and Combination." *ArXiv:1901.01751 [Cs, q-Fin, Stat]*, March. <http://arxiv.org/abs/1901.01751>.
- Krauss, Christopher, Xuan Anh Do, and Nicolas Huck. 2017. "Deep Neural Networks, Gradient-Boosted Trees, Random Forests: Statistical Arbitrage on the S&P 500." *European Journal of Operational Research* 259 (2): 689–702. <https://doi.org/10.1016/j.ejor.2016.10.031>.
- Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E Hinton. 2012. "ImageNet Classification with Deep Convolutional Neural Networks." In *Advances in Neural Information Processing Systems* 25, edited by F. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, 1097–1105. Curran Associates, Inc. <http://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf>.
- Lecun, Y., L. Bottou, Y. Bengio, and P. Haffner. 1998. "Gradient-Based Learning Applied to Document Recognition." *Proceedings of the IEEE* 86 (11): 2278–2324. <https://doi.org/10.1109/5.726791>.
- LeCun, Yann, Bernhard Boser, John S Denker, Donnie Henderson, Richard E Howard, Wayne Hubbard, and Lawrence D Jackel. 1989. "Backpropagation Applied to Handwritten Zip Code Recognition." *Neural Computation* 1 (4): 541–51.
- LeCun, Yann, Yoshua Bengio, and Geoffrey Hinton. 2015. "Deep Learning." *Nature* 521 (7553): 436–44. <https://doi.org/10.1038/nature14539>.
- Ledig, Christian, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, et al. 2017. "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network." In , 4681–90. http://openaccess.thecvf.com/content_cvpr_2017/html/Ledig_Photo-Realistic_Single_Image_CVPR_2017_paper.html.

- Ledoit, O., Wolf, M., 2003. *Improved estimation of the covariance matrix of stock returns with an application to portfolio selection*. *Journal of Empirical Finance* 10, 603–621. [https://doi.org/10.1016/S0927-5398\(03\)00007-0](https://doi.org/10.1016/S0927-5398(03)00007-0).
- Levy, Omer, and Yoav Goldberg. 2014. "Neural Word Embedding as Implicit Matrix Factorization." In *Advances in Neural Information Processing Systems 27*, edited by Z. Ghahramani, M. Welling, C. Cortes, N. D. Lawrence, and K. Q. Weinberger, 2177–2185. Curran Associates, Inc. <http://papers.nips.cc/paper/5477-neural-word-embedding-as-implicit-matrix-factorization.pdf>.
- Lin, Long-Ji, and Tom M Mitchell. 1992. *Memory Approaches to Reinforcement Learning in Non-Markovian Domains*. Citeseer.
- Liu, Yinhan, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. "RoBERTa: A Robustly Optimized BERT Pretraining Approach." *ArXiv:1907.11692 [Cs]*, July. <http://arxiv.org/abs/1907.11692>.
- Lo, A.W., 2002. *The Statistics of Sharpe Ratios*. <https://doi.org/10.2469/faj.v58.n4.2453>.
- Maaten, Laurens van der, and Geoffrey Hinton. 2008. "Visualizing Data Using T-SNE." *Journal of Machine Learning Research* 9 (Nov): 2579–2605. <http://www.jmlr.org/papers/v9/vandermaaten08a.html>.
- Madhavan, Ananth. 2002. "Market Microstructure: A Practitioner's Guide." *Financial Analysts Journal* 58 (5): 28–42. www.jstor.org/stable/4480415.
- Madhavan, Ananth. 2000. "Market Microstructure: A Survey." *Journal of Financial Markets* 3 (3): 205–58. [https://doi.org/10.1016/S1386-4181\(00\)00007-0](https://doi.org/10.1016/S1386-4181(00)00007-0).
- Malkiel, Burton G. 2003. "The Efficient Market Hypothesis and Its Critics." *Journal of Economic Perspectives* 17 (1): 59–82. <https://doi.org/10.1257/089533003321164958>.
- Man, Xiliu, Tong Luo, and Jianwu Lin. 2019. "Financial Sentiment Analysis(FSA): A Survey." In *2019 IEEE International Conference on Industrial Cyber Physical Systems (ICPS)*, 617–22. <https://doi.org/10.1109/ICPHYS.2019.8780312>.
- Markowitz, H., 1952. *Portfolio Selection*. *The Journal of Finance* 7, 77–91. <https://doi.org/10.2307/2975974>.
- Meredith, Mike, and John Kruschke. 2018. "Bayesian Estimation Supersedes the T-Test," 14.
- Michaud, Richard O., Esch, D.N., Michaud, Robert O., 2017. *The "Fundamental Law of Active Management" is No Law of Anything*. <https://doi.org/10.2139/ssrn.2834020>.
- Mikolov, Tomas, Ilya Sutskever, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. "Distributed Representations of Words and Phrases and Their Compositionality." In *Proceedings of the 26th International Conference on Neural Information Processing Systems – Volume 2*, 3111–3119. NIPS'13. USA: Curran Associates Inc. <http://dl.acm.org/citation.cfm?id=2999792.2999959>.
- Miller, Rena S. 2016. "High Frequency Trading: Overview of Recent Developments." *High Frequency Trading*, 19.

- Mimno, David, Hanna M. Wallach, Edmund Talley, Miriam Leenders, and Andrew McCallum. 2011. "Optimizing Semantic Coherence in Topic Models." In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 262–272. EMNLP '11. Edinburgh, United Kingdom: Association for Computational Linguistics.
- Mitchell, Tom M. 1997. "Machine Learning."
- Mnih, Andriy, and Koray Kavukcuoglu. 2013. "Learning Word Embeddings Efficiently with Noise-Contrastive Estimation." In *Advances in Neural Information Processing Systems 26*, edited by C. J. C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Q. Weinberger, 2265–2273. Curran Associates, Inc. <http://papers.nips.cc/paper/5165-learning-word-embeddings-efficiently-with-noise-contrastive-estimation.pdf>.
- Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Alex Graves, Ioannis Antonoglou, Daan Wierstra, and Martin Riedmiller. 2013. "Playing Atari with Deep Reinforcement Learning." *ArXiv:1312.5602 [Cs]*, December. <http://arxiv.org/abs/1312.5602>.
- Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, et al. 2015. "Human-Level Control through Deep Reinforcement Learning." *Nature* 518 (7540): 529–33. <https://doi.org/10.1038/nature14236>.
- Morin, Frederic, and Yoshua Bengio. 2005. "Hierarchical Probabilistic Neural Network Language Model." In *Aistats*, 5:246–52.
- Nakamoto, Yukikazu. 2011. "A Short Introduction to Learning to Rank." *IEICE Transactions on Information and Systems* E94-D (1): 1–2. <https://doi.org/10.1587/transinf.E94.D.1>.
- Nasseri, Alya Al, Allan Tucker, and Sergio de Cesare. 2015. "Quantifying StockTwits Semantic Terms' Trading Behavior in Financial Markets: An Effective Application of Decision Tree Algorithms." *Expert Systems with Applications* 42 (23): 9192–9210. <https://doi.org/10.1016/j.eswa.2015.08.008>.
- Netzer, Yuval. 2011. "Reading Digits in Natural Images with Unsupervised Feature Learning." In *NIPS Workshop on Deep Learning*.
- Nevmyvaka, Yuriy, Yi Feng, and Michael Kearns. 2006. "Reinforcement Learning for Optimized Trade Execution." In *Proceedings of the 23rd International Conference on Machine Learning*, 673–680. ICML '06. Pittsburgh, Pennsylvania, USA: Association for Computing Machinery. <https://doi.org/10.1145/1143844.1143929>.
- Ng, Andrew Y., and Michael I. Jordan. 2002. "On Discriminative vs. Generative Classifiers: A Comparison of Logistic Regression and Naive Bayes." In *Advances in Neural Information Processing Systems 14*, edited by T. G. Dietterich, S. Becker, and Z. Ghahramani, 841–848. MIT Press. <http://papers.nips.cc/paper/2020-on-discriminative-vs-generative-classifiers-a-comparison-of-logistic-regression-and-naive-bayes.pdf>.

- Nilsson, Nils J. 2009. *The Quest for Artificial Intelligence*. Cambridge University Press.
- Novy-Marx, R., 2015. *Backtesting Strategies Based on Multiple Signals (Working Paper No. 21329)*. National Bureau of Economic Research. <https://doi.org/10.3386/w21329>.
- Odena, Augustus. 2019. "Open Questions about Generative Adversarial Networks." *Distill* 4 (4): e18. <https://doi.org/10.23915/distill.00018>.
- Pan, Zhaoqing, Weijie Yu, Xiaokai Yi, Asifullah Khan, Feng Yuan, and Yuhui Zheng. 2019. "Recent Progress on Generative Adversarial Networks (GANs): A Survey." *IEEE Access* 7: 36322–33. <https://doi.org/10.1109/ACCESS.2019.2905015>.
- Pennington, Jeffrey, Richard Socher, and Christopher Manning. 2014. "Glove: Global Vectors for Word Representation." In *EMNLP*, 14:1532–43. <https://doi.org/10.3115/v1/D14-1162>.
- Perold, A.F., 2004. *The Capital Asset Pricing Model*. *Journal of Economic Perspectives* 18, 3–24. <https://doi.org/10.1257/0895330042162340>.
- Prabhavalkar, Rohit, Kanishka Rao, Tara N. Sainath, Bo Li, Leif Johnson, and Navdeep Jaitly. 2017. "A Comparison of Sequence-to-Sequence Models for Speech Recognition." In *Interspeech 2017*, 939–43. ISCA. <https://doi.org/10.21437/Interspeech.2017-233>.
- Prado, M.L. de, 2016. "Building Diversified Portfolios that Outperform Out of Sample." *The Journal of Portfolio Management* 42, 59–69. <https://doi.org/10.3905/jpm.2016.42.4.059>.
- Preis, Tobias, Helen Susannah Moat, and H. Eugene Stanley. 2013. "Quantifying Trading Behavior in Financial Markets Using Google Trends." *Scientific Reports* 3 (1): 1–6. <https://doi.org/10.1038/srep01684>.
- Prokhorenkova, Liudmila, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2019. "CatBoost: Unbiased Boosting with Categorical Features." *ArXiv:1706.09516 [Cs]*, January. <http://arxiv.org/abs/1706.09516>.
- Radford, Alec, Luke Metz, and Soumith Chintala. 2016. "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks." *ArXiv:1511.06434 [Cs]*, January. <http://arxiv.org/abs/1511.06434>.
- Raffinot, T., 2017. *Hierarchical Clustering-Based Asset Allocation*. *The Journal of Portfolio Management* 44, 89–99. <https://doi.org/10.3905/jpm.2018.44.2.089>.
- Rasekhschaffe, Keywan Christian, and Robert C. Jones. 2019. "Machine Learning for Stock Selection." *Financial Analysts Journal* 75 (3): 70–88. <https://doi.org/10.1080/0015198X.2019.1596678>.
- Redmon, Joseph. 2016. "You Only Look Once: Unified, Real-Time Object Detection." *ArXiv:1506.02640 [Cs]*, May.
- Reed, Scott, Zeynep Akata, Santosh Mohan, Samuel Tenka, Bernt Schiele, and Honglak Lee. 2016. "Learning What and Where to Draw." *ArXiv:1610.02454 [Cs]*, October. <http://arxiv.org/abs/1610.02454>.

- Reinganum, Marc R. 1981. "Misspecification of Capital Asset Pricing: Empirical Anomalies Based on Earnings' Yields and Market Values." *Journal of Financial Economics* 9 (1): 19–46.
- Ren, Shaoqing, Kaiming He, Ross Girshick, and Jian Sun. 2015. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks." In *Advances in Neural Information Processing Systems* 28, edited by C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, 91–99. Curran Associates, Inc. <http://papers.nips.cc/paper/5638-faster-r-cnn-towards-real-time-object-detection-with-region-proposal-networks.pdf>.
- Roa-Vicens, Jacobo, Cyrine Chtourou, Angelos Filos, Francisco Rullan, Yarin Gal, and Ricardo Silva. 2019. "Towards Inverse Reinforcement Learning for Limit Order Book Dynamics." *ArXiv:1906.04813 [Cs, q-Fin, Stat]*, June. <http://arxiv.org/abs/1906.04813>.
- Röder, Michael, Andreas Both, and Alexander Hinneburg. 2015. "Exploring the Space of Topic Coherence Measures." In *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining*, 399–408. WSDM '15. Shanghai, China: Association for Computing Machinery. <https://doi.org/10.1145/2684822.2685324>.
- Rokach, Lior, and Oded Z. Maimon. 2008. *Data Mining with Decision Trees: Theory and Applications*. World Scientific.
- Roll, Richard, and Stephen A. Ross. 1984. "The Arbitrage Pricing Theory Approach to Strategic Portfolio Planning." *Financial Analysts Journal* 40 (3): 14–26. <https://doi.org/10.2469/faj.v40.n3.14>.
- Romero, P.J., Balch, T., 2014. *What Hedge Funds Really Do: An Introduction to Portfolio Management*. Business Expert Press.
- Ruder, Sebastian. 2017. "An Overview of Gradient Descent Optimization Algorithms." *ArXiv:1609.04747 [Cs]*, June. <http://arxiv.org/abs/1609.04747>.
- Salimans, Tim, Diederik P. Kingma, and Max Welling. 2015. "Markov Chain Monte Carlo and Variational Inference: Bridging the Gap." *ArXiv:1410.6460 [Stat]*, May. <http://arxiv.org/abs/1410.6460>.
- Samuelson, P., Thorp, E., T. Kassouf, S., 1968. *Beat the Market: A Scientific Stock Market System*. *Journal of the American Statistical Association* 63, 1049. <https://doi.org/10.2307/2283900>.
- Saul, Lawrence K, and Sam T Roweis. 2000. "An Introduction to Locally Linear Embedding." Unpublished. Available at: <https://cs.nyu.edu/~roweis/lle/papers/lleintro.pdf>.
- Schapire, Robert E., and Yoav Freund. 2012. *Boosting: Foundations and Algorithms*. MIT Press.
- Schaul, Tom, John Quan, Ioannis Antonoglou, and David Silver. 2015. "Prioritized Experience Replay." *ArXiv:1511.05952 [Cs]*, November. <http://arxiv.org/abs/1511.05952>.

- Schuster, M., and K.K. Paliwal. 1997. "Bidirectional Recurrent Neural Networks." *IEEE Transactions on Signal Processing* 45 (11): 2673–81. <https://doi.org/10.1109/78.650093>.
- Sezer, Omer Berat, and Ahmet Murat Ozbayoglu. 2018. "Algorithmic Financial Trading with Deep Convolutional Neural Networks: Time Series to Image Conversion Approach." *Applied Soft Computing* 70 (September): 525–38. <https://doi.org/10.1016/j.asoc.2018.04.024>.
- Sievert, Carson, and Kenneth Shirley. 2014. "LDAvis: A Method for Visualizing and Interpreting Topics." In *Proceedings of the Workshop on Interactive Language Learning, Visualization, and Interfaces*, 63–70.
- Sigtia, Siddharth, Emmanouil Benetos, Srikanth Cherla, Tillman Weyde, A. Garcez, and Simon Dixon. 2014. "RNN-Based Music Language Models for Improving Automatic Music Transcription."
- Simonyan, Karen. 2015. "Very Deep Convolutional Networks for Large-Scale Image Recognition." *ArXiv:1409.1556 [Cs]*, April.
- Srivastava, Nitish, Elman Mansimov, and Ruslan Salakhutdinov. 2016. "Unsupervised Learning of Video Representations Using LSTMs," 10.
- Stevens, Keith, Philip Kegelmeyer, David Andrzejewski, and David Buttler. 2012. "Exploring Topic Coherence over Many Models and Many Topics."
- Strumeyer, Gary. 2017. *The Capital Markets: Evolution of the Financial Ecosystem*. John Wiley & Sons.
- Sutskever, Ilya, James Martens, George Dahl, and Geoffrey Hinton. 2013. "On the Importance of Initialization and Momentum in Deep Learning." In *International Conference on Machine Learning*, 1139–47. <http://proceedings.mlr.press/v28/sutskever13.html>.
- Sutton, Richard S, and Andrew G Barto. 2018. *Reinforcement Learning: An Introduction*. MIT press.
- Sutton, Richard S, David A. McAllester, Satinder P. Singh, and Yishay Mansour. 2000. "Policy Gradient Methods for Reinforcement Learning with Function Approximation." In *Advances in Neural Information Processing Systems* 12, edited by S. A. Solla, T. K. Leen, and K. Müller, 1057–1063. MIT Press. <http://papers.nips.cc/paper/1713-policy-gradient-methods-for-reinforcement-learning-with-function-approximation.pdf>.
- Szegedy, Christian, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. 2015. "Going Deeper with Convolutions." In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1–9. <https://doi.org/10.1109/CVPR.2015.7298594>.
- Trichilo, D., Braun, J.L., 2005. *Extending the Fundamental Law of Investment Management*.

- Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. "Attention Is All You Need." In *Advances in Neural Information Processing Systems* 30, edited by I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, 5998–6008. Curran Associates, Inc. <http://papers.nips.cc/paper/7181-attention-is-all-you-need.pdf>.
- Vincent, Pascal, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. 2008. "Extracting and Composing Robust Features with Denoising Autoencoders." In *Proceedings of the 25th International Conference on Machine Learning*, 1096–1103. ICML '08. Helsinki, Finland: Association for Computing Machinery. <https://doi.org/10.1145/1390156.1390294>.
- Wang, Alex, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2019. "SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems." *ArXiv:1905.00537 [Cs]*, May. <http://arxiv.org/abs/1905.00537>.
- Wang, Jia, Tong Sun, Benyuan Liu, Yu Cao, and Hongwei Zhu. 2019. "CLVSA: A Convolutional LSTM Based Variational Sequence-to-Sequence Model with Attention for Predicting Trends of Financial Markets." In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, 3705–11. Macao, China: International Joint Conferences on Artificial Intelligence Organization. <https://doi.org/10.24963/ijcai.2019/514>.
- Watkins, Christopher J. C. H., and Peter Dayan. 1992. "Q-Learning." In *Machine Learning*, 279–292.
- Watkins, Christopher John Cornish Hellaby. 1989. "Learning from Delayed Rewards.", http://www.cs.rhul.ac.uk/~chrisw/new_thesis.pdf.
- Werbos, P.J. 1990. "Backpropagation through Time: What It Does and How to Do It." *Proceedings of the IEEE* 78 (10): 1550–60. <https://doi.org/10.1109/5.58337>.
- Wiese, Magnus, Robert Knobloch, Ralf Korn, and Peter Kretschmer. 2019. "Quant GANs: Deep Generation of Financial Time Series." *ArXiv:1907.06673 [Cs, q-Fin, Stat]*, December. <http://arxiv.org/abs/1907.06673>.
- Williams, Ronald J, and David Zipser. 1989. "A Learning Algorithm for Continually Running Fully Recurrent Neural Networks." *Neural Computation* 1 (2): 270–80.
- Wooldridge, Jeffrey M. 2002. "Econometric Analysis of Cross Section and Panel Data MIT Press." *Cambridge*, MA 108.
- Wooldridge, Jeffrey M. 2008. *Introductory Econometrics: A Modern Approach*. Cengage Learning.
- Yoon, Jinsung, Daniel Jarrett, and Mihaela van der Schaar. 2019. "Time-Series Generative Adversarial Networks." In *Advances in Neural Information Processing Systems* 32, edited by H. Wallach, H. Larochelle, A. Beygelzimer, F. d\textquotesingle Alché-Buc, E. Fox, and R. Garnett, 5508–5518. Curran Associates, Inc. <http://papers.nips.cc/paper/8789-time-series-generative-adversarial-networks.pdf>.

- Zhang, Han, Tao Xu, Hongsheng Li, Shaoting Zhang, Xiaogang Wang, Xiaolei Huang, and Dimitris Metaxas. 2017. "StackGAN: Text to Photo-Realistic Image Synthesis with Stacked Generative Adversarial Networks." *ArXiv:1612.03242 [Cs, Stat]*, August. <http://arxiv.org/abs/1612.03242>.
- Zhou, Xingyu, Zhisong Pan, Guyu Hu, Siqi Tang, and Cheng Zhao. 2018. "Stock Market Prediction on High-Frequency Data Using Generative Adversarial Nets." *Research Article: Mathematical Problems in Engineering. Hindawi*. 2018. <https://doi.org/10.1155/2018/4907423>.
- Zhu, Jun-Yan, Taesung Park, Phillip Isola, and Alexei A. Efros. 2018. "Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks." *ArXiv:1703.10593 [Cs]*, November. <http://arxiv.org/abs/1703.10593>.

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